unit 5 · Graph (Original)

Title: Social Networking Application using Graph.

problem statement:

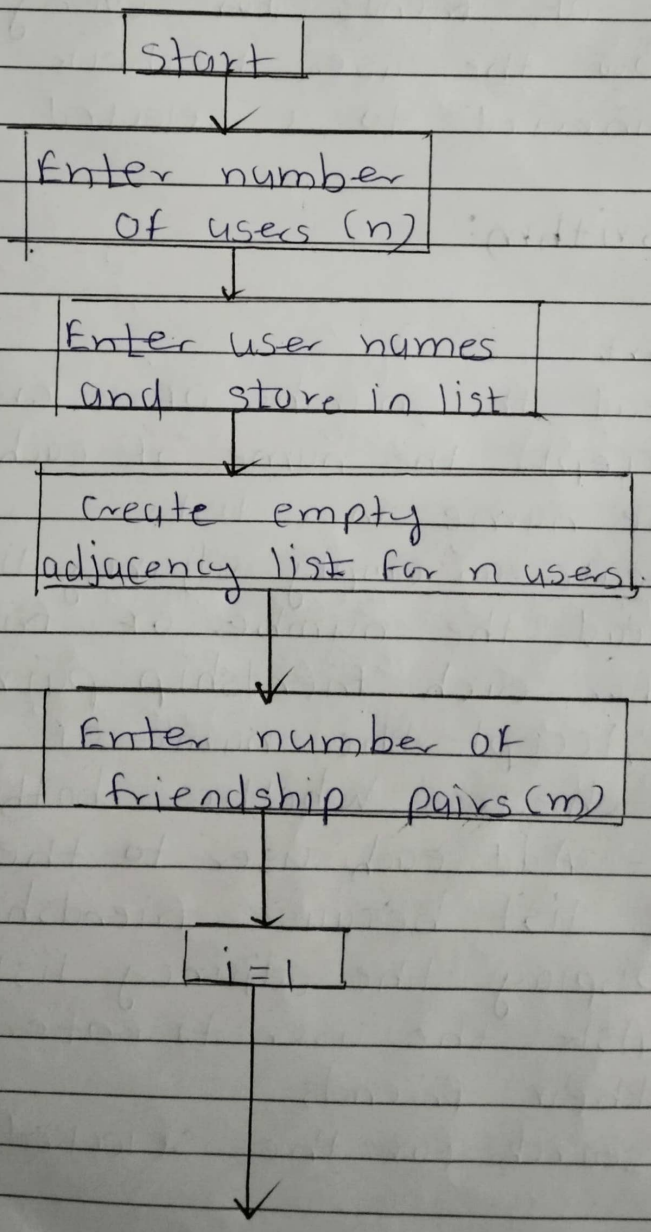
Develop a c++ program to represent users of a social networking application as vertices and friendships as undirected edges. Implement the graph using an adjancy list. Accept user friedship pairs, display the list of friends for every user, and allow the user to check the friends direclty connected to a selected user.

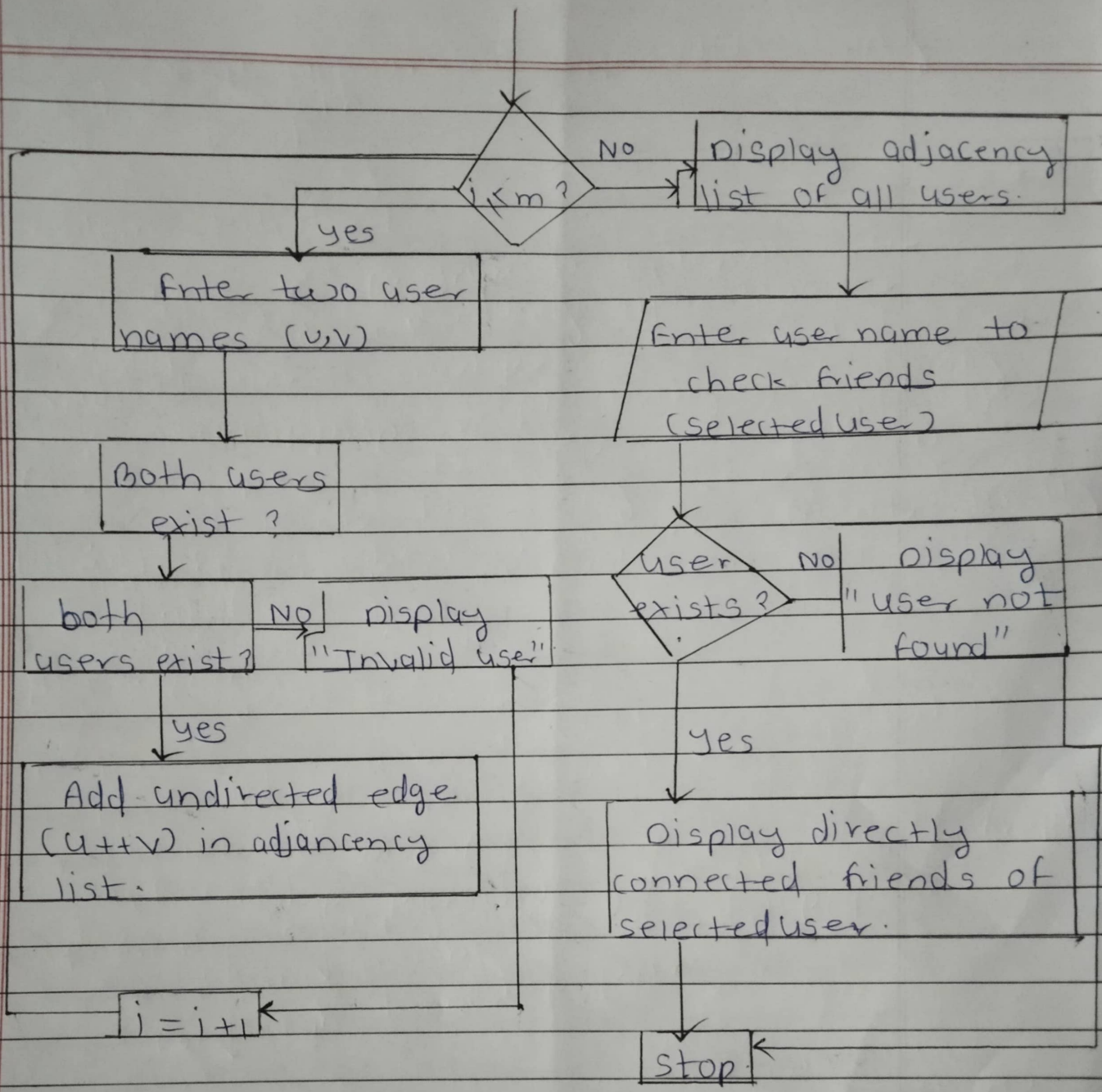
Algorithm:

1. Start.
2. Read the number of users.
3. Accept the name of each user and store the name in a list.
4. Create an empty adjancy list for the graph.
5. Read the number of friendship pairs.
6. For each friendship pair.
   - Accept the name of two users.
   - checks whether both users exists.
   - Add each user to the other's adjancy list because friendship is undirected.
7. Display the adjancy list of all user.
8. Ask the user to enter a name to check their friends.
9. search for the selected user.

10. If the user exists, display all directly connected friends.
Otherwise, display an appropriate error message.
11. Stop.

Flowchart:




i<m ?
No
Display adjacency list of all users.
yes
Enter two user names (u,v)
Enter user name to check friends (selected user)
Both users exist ?
User exists ?
No
Display "user not found"
both users exist ?
NO
Display "Invalid user"
yes
yes
Add undirected edge (u++v) in adjancency list.
Display directly connected friends of selected user.
i=i+1
Stop

```
include <iostream>
#include <vector>
#include <string>
#include <unordered_map>
using namespace std;

int main()
{
   int n, m;

   cout << "Enter number of users: ";
   cin >> n;

   if (n <= 0)
   {
      cout << "Number of users must be positive.\n";
      return 0;
   }

   vector<string> users(n);
   unordered_map<string, int> userIndex;

   cout << "Enter user names:\n";

   for (int i = 0; i < n; i++)
   {
      cin >> users[i];
```

```
    if (userIndex.count(users[i]))
    {
        cout << "Duplicate user name not allowed.\n";
        return 0;
    }

    userIndex[users[i]] = i;
}

vector<vector<int>> adj(n);

cout << "Enter number of friendship pairs: ";
cin >> m;

if (m < 0)
{
    cout << "Number of friendships cannot be negative.\n";
    return 0;
}

cout << "Enter friendship pairs:\n";

for (int i = 0; i < m; i++)
{
    string u, v;
    cin >> u >> v;

    if (!userIndex.count(u) || !userIndex.count(v))
```

```
    {
        cout << "Invalid user in friendship pair.\n";
        continue;
    }

    int a = userIndex[u];
    int b = userIndex[v];

    if (a == b)
    {
        cout << "Self-friendship is not allowed.\n";
        continue;
    }

    adj[a].push_back(b);
    adj[b].push_back(a);
}

cout << "\nAdjacency List:\n";

for (int i = 0; i < n; i++)
{
    cout << users[i] << " -> ";

    if (adj[i].empty())
    {
        cout << "No friends";
    }
```

```
    else
    {
        for (int friendIndex : adj[i])
        {
            cout << users[friendIndex] << " ";
        }
    }

    cout << "\n";
}

string selectedUser;
cout << "\nEnter user name to check friends: ";
cin >> selectedUser;

if (userIndex.count(selectedUser))
{
    int index = userIndex[selectedUser];

    cout << "Friends of " << selectedUser << ": ";

    if (adj[index].empty())
    {
        cout << "No friends";
    }
    else
    {
        for (int friendIndex : adj[index])
```

```
            {
                cout << users[friendIndex] << " ";
            }
        }

        cout << "\n";
    }
    else
    {
        cout << "User not found.\n";
    }

    return 0;
}
```


File Edit Selection View Go Run Terminal Help
c programing code
Problems Output Debug Console Terminal Ports
powershell
PS D:\c programing code> g++ 4.code.cpp
PS D:\c programing code> ./a.exe
Enter number of users: 3
Enter user names:
ashish shubham yash
Enter number of friendship pairs: 2
Enter friendship pairs:
ashish shubham
shubham yash
Enter new user name: shiddhant
Enter friend name: ashish
User and friendship added successfully.
Updated Adjacency List:
ashish -> shubham shiddhant
shubham -> ashish yash
yash -> shubham
shiddhant -> ashish
PS D:\c programing code>
BLACKBOX Agent
Open Website
Ln 109, Col 1
Spaces: 4
UTF-8
CRLF
C++
Go Live
AI Code Chat
Win32
Search
ENG
IN
60%
17:21
25-10-2026